

18

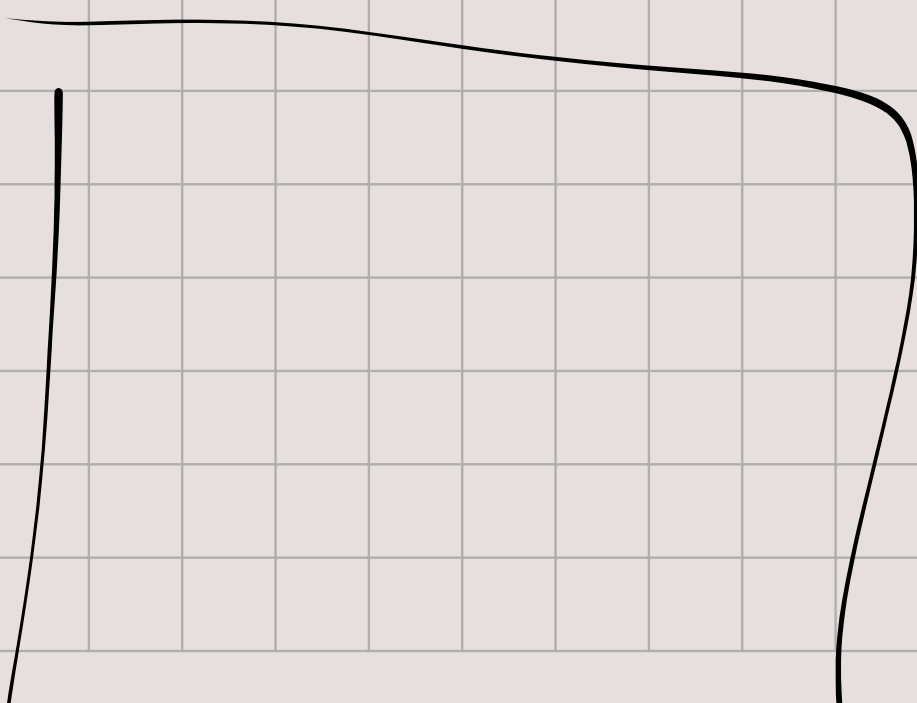
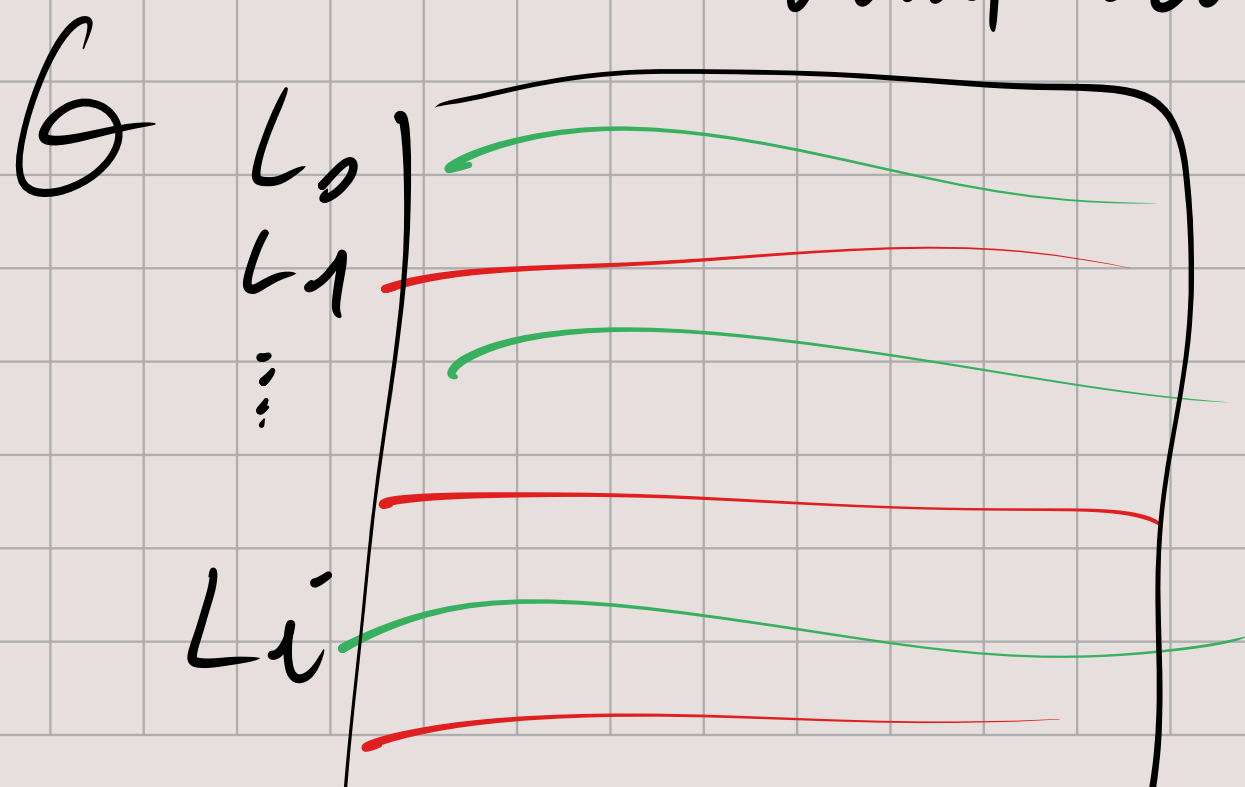
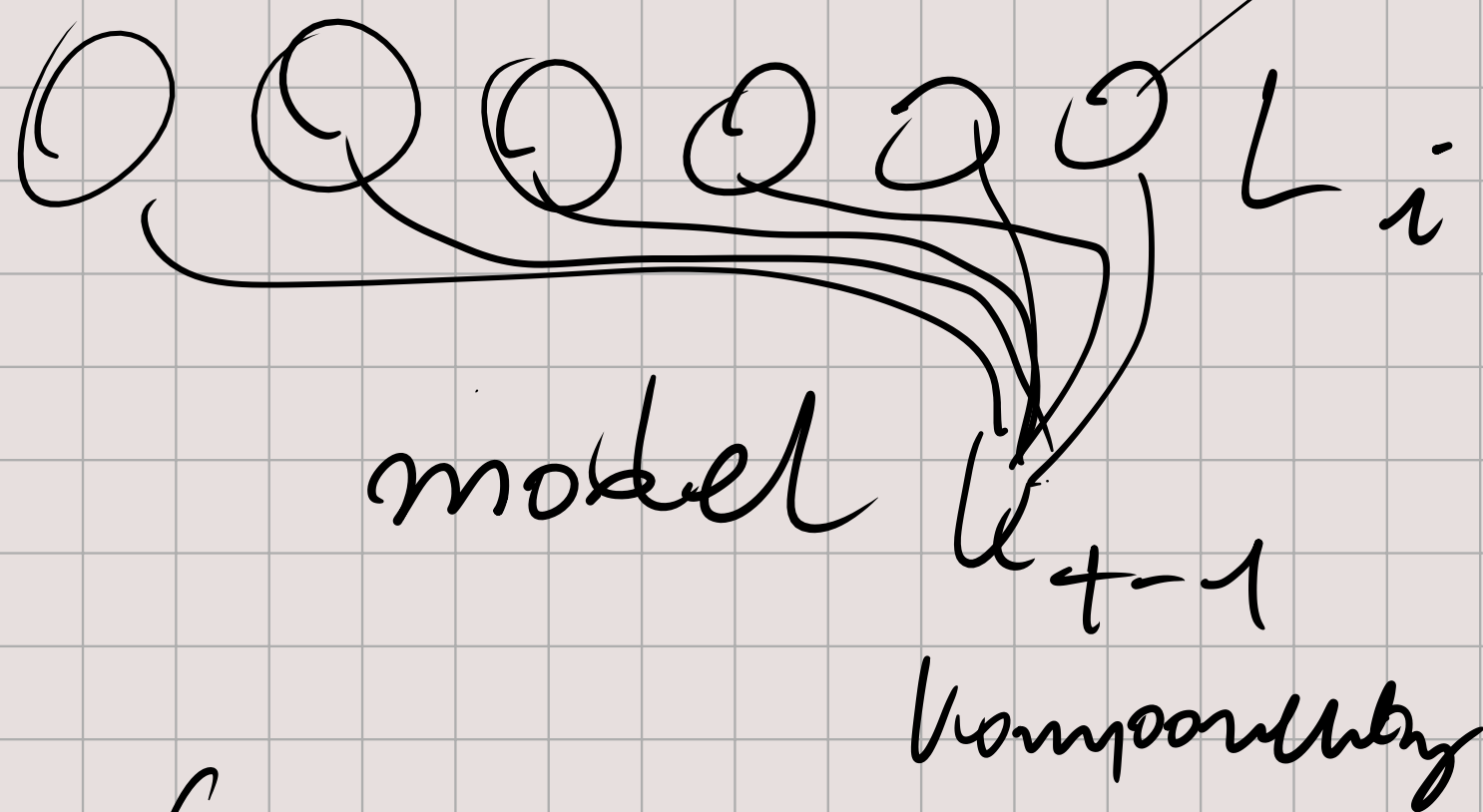
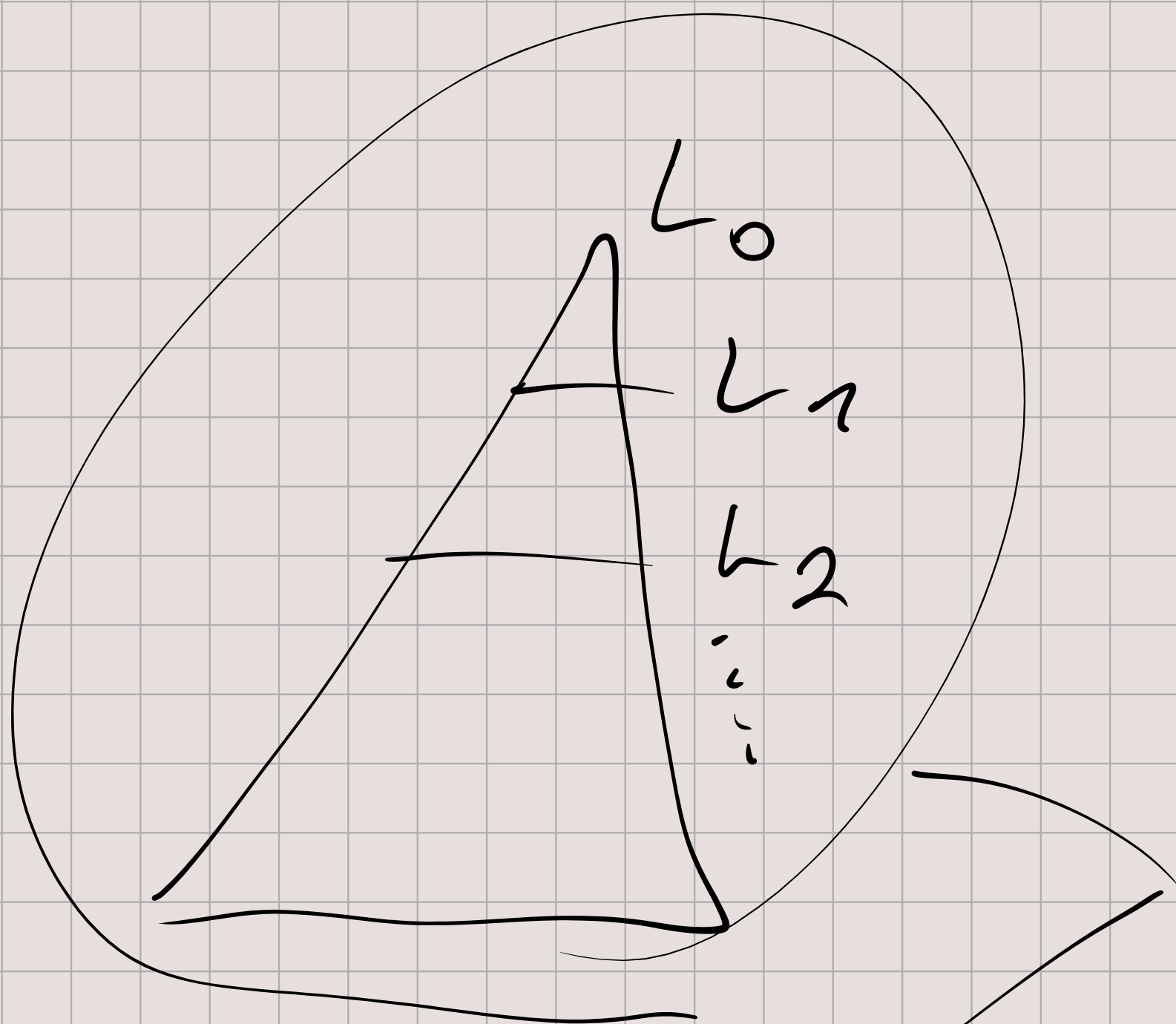
G spójny
 $\{L_i \mid i \in \mathbb{N}\}$

bo jak pokazać dla
dowolnego komponentu
to będzie prawdziwe dla
dowolnego grafu.

$$L_0 = \{v\}$$

$$L_i = \{u \mid \text{dist}_G(L_0, u) = i\}$$

$$\forall i, K_t \triangleleft G \Rightarrow K_{t-1} \triangleleft G[L_i]$$



Czerwone nie mają

$K_2 \nsubseteq G \Leftrightarrow G$ nie ma krawędzi

$$\Leftrightarrow \chi(G) \leq 1$$

15

G graf $t \in \mathbb{N}$

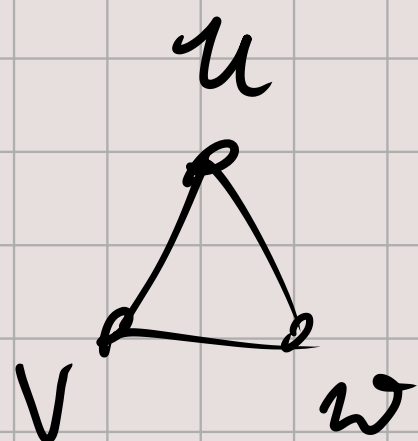
$$|V(G)| \leq t^2, \quad \delta(G) \geq \frac{t^2}{2}$$

$$\Rightarrow K_{\lfloor ct \rfloor} \subset G$$

$$H: V(H) = V(G)$$

$$u, v \in E(H) \Rightarrow |N_G(u) \cap N_G(v)| \leq \frac{t^2}{6}$$

H nie ma Δ



$$|N(v) - N(u)| \leq \frac{t^2}{3}$$

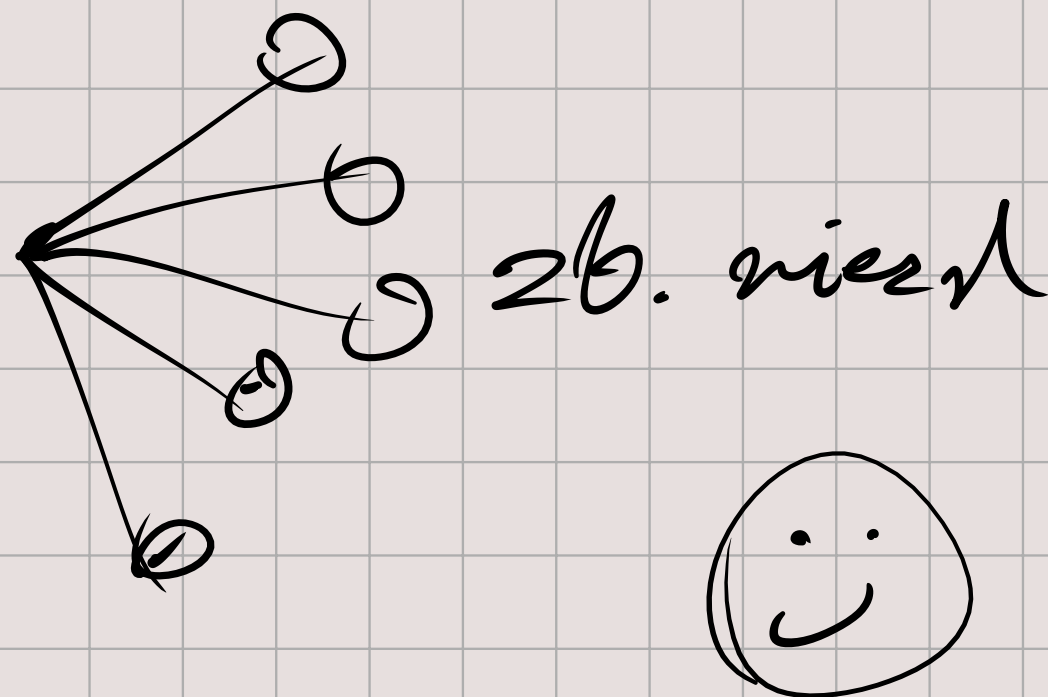
$$|N(w) - N(u)| \leq \frac{t^2}{3}$$

$$\Rightarrow |N(v) \cap N(u)| \geq \frac{t^2}{6}$$

bo u nie jest
w sąsiedztwie

Teza H ma zb. niezależny wielkości $\frac{1}{2}t$

$$\exists v \in V(H) \deg_H(v) \geq t$$



$$\text{wpp. } \Delta(H) < t, \chi(H) \leq t$$

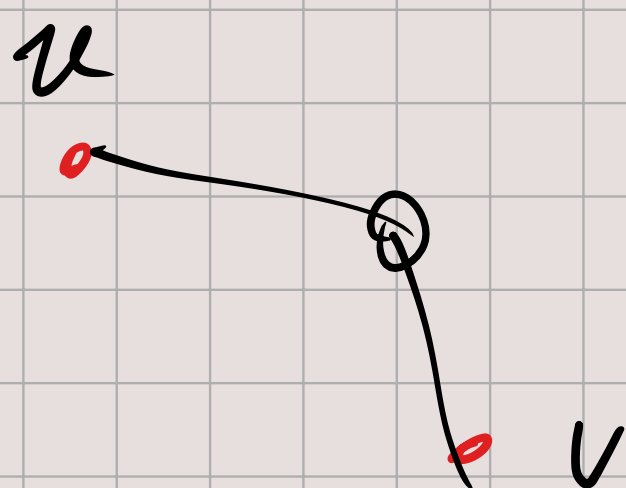
$$\alpha(H) \cdot \chi(H) \geq n(H) := |V(H)|$$

bo każdy kolor mi daje niezależny zb.

$$\alpha(H) \geq \frac{n(H)}{\chi(H)} \geq \frac{\frac{1}{2}t^2}{t} = \frac{1}{2}t$$

I zb. niezal w H

$$|I| \sim \frac{t}{\sqrt{6}}$$



wtedy topologizmie podprzestrzeni

zda więc zamodelować K_t

[16]

go tych sąsiadów, między którymi prowadzi sieć jest wyst. dużo.

$$(19) \quad k_t < G \Rightarrow K \leq ct?$$

$$\text{od}(G) \geq \alpha t^2 \Rightarrow k_t < G$$

$$\exists H \subseteq G \quad \delta(H) \geq \frac{1}{2} \alpha t^2$$

nie działa

2 pop. zestawu

$$G \quad 12 = \text{zest. 9}$$

$$\exists H \subseteq G \quad \delta(H) \geq \frac{\text{od}(G)}{2}$$

ustawiamy wielkość

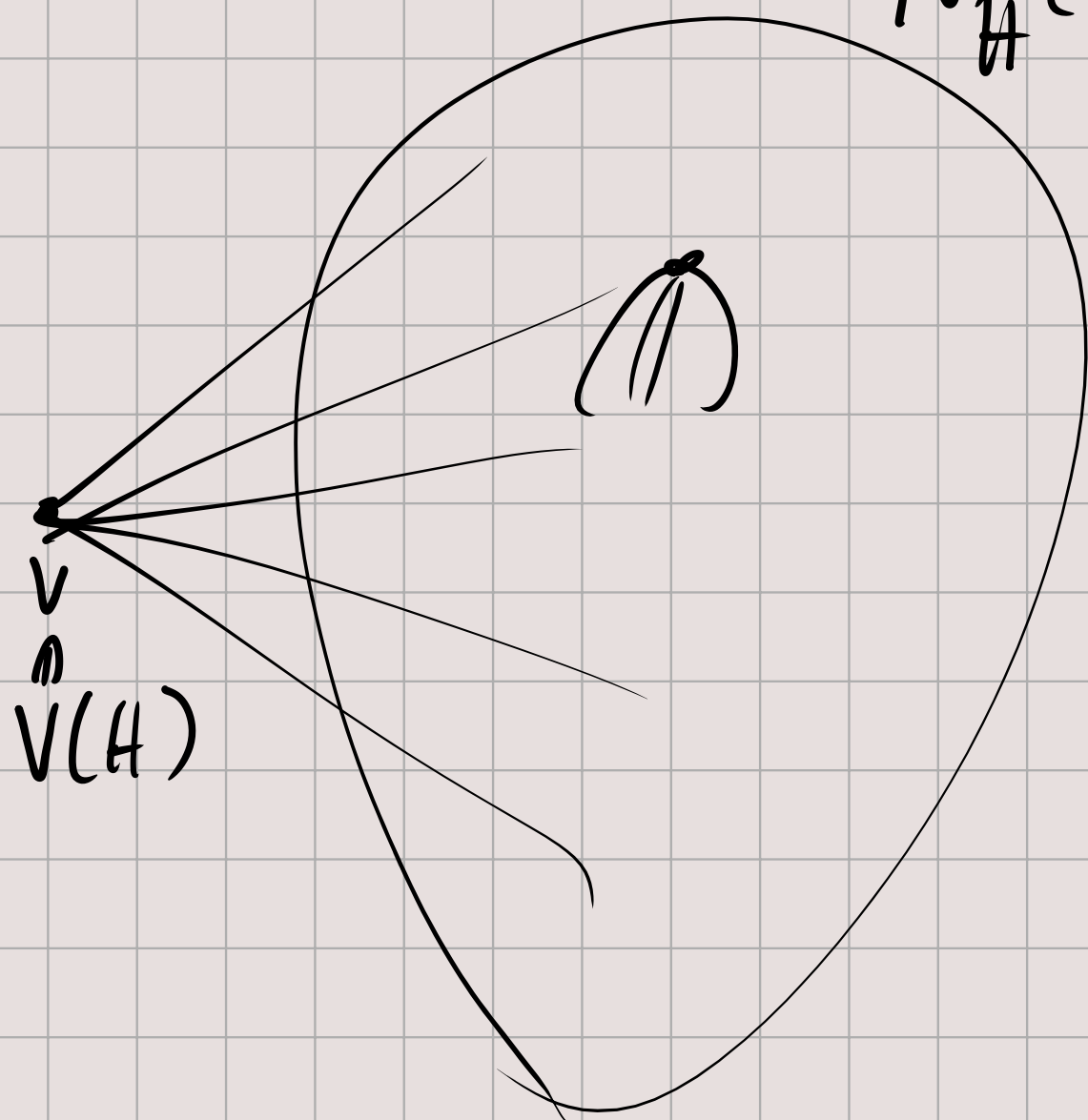
$$\text{t.j. } \deg v < \frac{\text{od}(G)}{2}$$

$H < G$ minimalny

$$\text{od}(H) \geq \alpha t^2 \quad \text{minimalny}$$

$\exists$ i max co najmniej jedną krawędź

$N_H(v)$

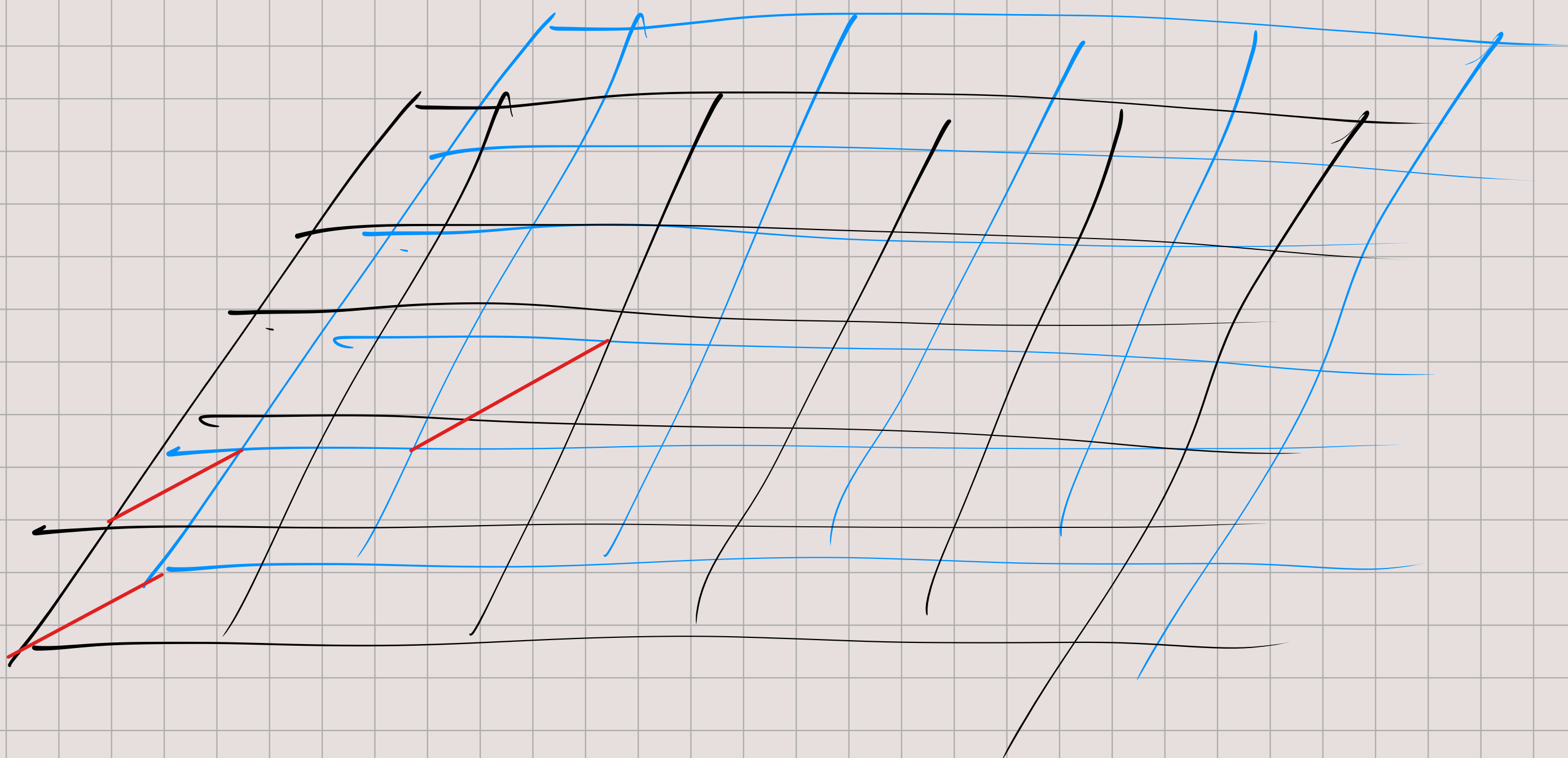


$$\delta \geq t^2$$

może dokonać? $H \cup H$

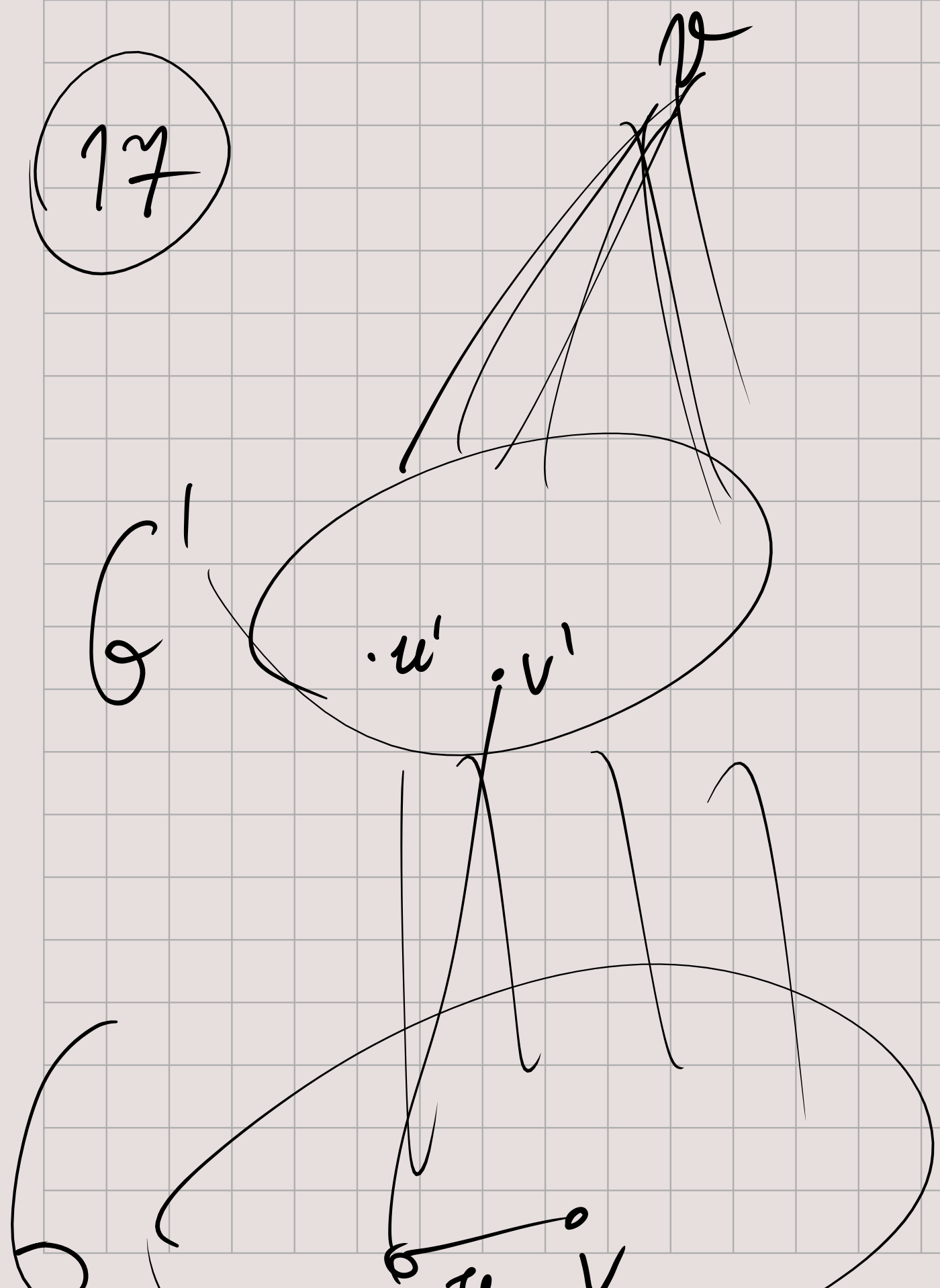
16

$$[2025] \times [2025] \times [2]$$

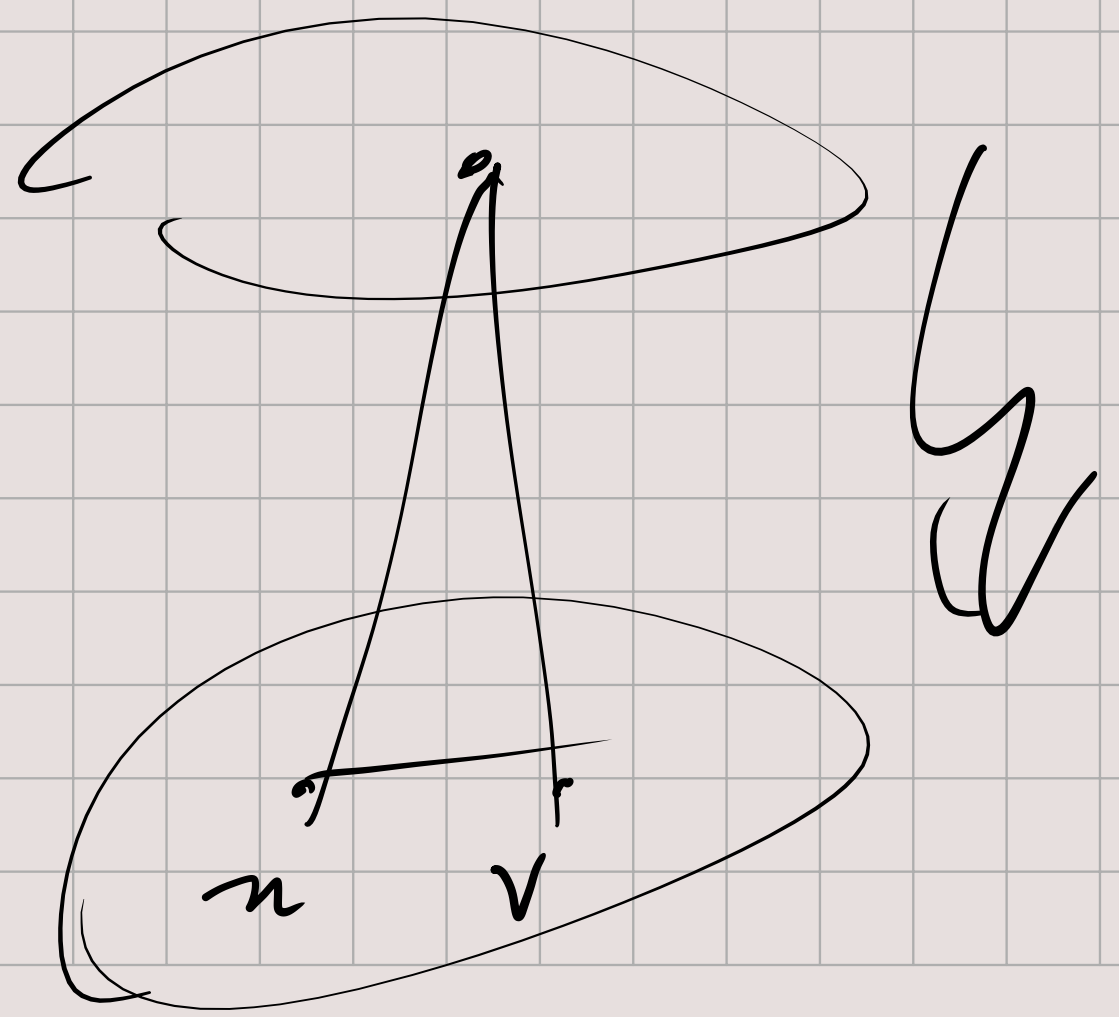


na jednej płaszczyźnie znajduje k_4
 jako minor a potem konstruuje
 cały górę w jeden wielokąt.

17



1 Niech Δ



$$\chi(G_{n+1}) \geq n$$

By using induction

$$\text{t. i.e. } c(v) = n$$

$$c : G_{n+1} \rightarrow [n]$$

$$\downarrow$$

$$d : G_n \rightarrow [n-1]$$

if $v \in G_n$ i $c(v) \neq n$ t $d(v) = c(v)$

if $c(v) = n$, to $d(v) = c(v')$

$$u, v \in G_n \quad uv \in G_n$$

$$(1) \quad c(u) = c(v) \quad \checkmark$$

$$(2) \quad c(u) \neq c(v) \quad \checkmark$$

$$\quad \quad \quad \times \quad \times$$

$$\quad \quad \quad n$$